

The Leaf

Leaf Anatomy

A typical leaf has a **leaf base** (may bear stipules, form sheath in monocots, or pulvinus in legumes), a **petiole**, and the **lamina** (leaf blade) with the midrib.

Venation

Reticulate venation (net-like, dicots) vs **parallel venation** (veins run parallel, monocots).

Simple vs Compound Leaves

In a **simple leaf** incisions don't reach the midrib. In a **compound leaf** incisions reach the midrib, forming leaflets — **pinnately compound** (Neem) or **palmately compound** (Silk Cotton).

Phyllotaxy

The pattern of leaf arrangement: **alternate** (China rose), **opposite** (Calotropis), and **whorled** (Alstonia).

Leaf Modifications

Leaves can modify into **tendrils** for climbing (Pea), **spines** for protection (Cactus, reducing water loss), and **leaf pitchers** (Nepenthes) for insect capture — showcasing structural plasticity for varied ecological roles.

Heterophylly

Some plants show **heterophylly**, producing leaves of different shapes at different developmental stages or environmental conditions (e.g., Larkspur, cotton), an adaptive response reflecting varying functional needs.

Types of Phyllotaxy

Type	Arrangement	Example
Alternate	One leaf per node, alternating	China rose, Mustard, Sunflower
Opposite	Pair of leaves per node	Calotropis, Guava
Whorled	More than 2 leaves per node	Alstonia

3 Types of Phyllotaxy

- Alternate
- Opposite
- Whorled